

Basic Laws and Analysis

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Kirchoff's Laws in the Frequency domain

- KVL:

Kirchhoff's voltage law (KVL) states that the algebraic sum of all voltages around a closed path (or loop) is zero.

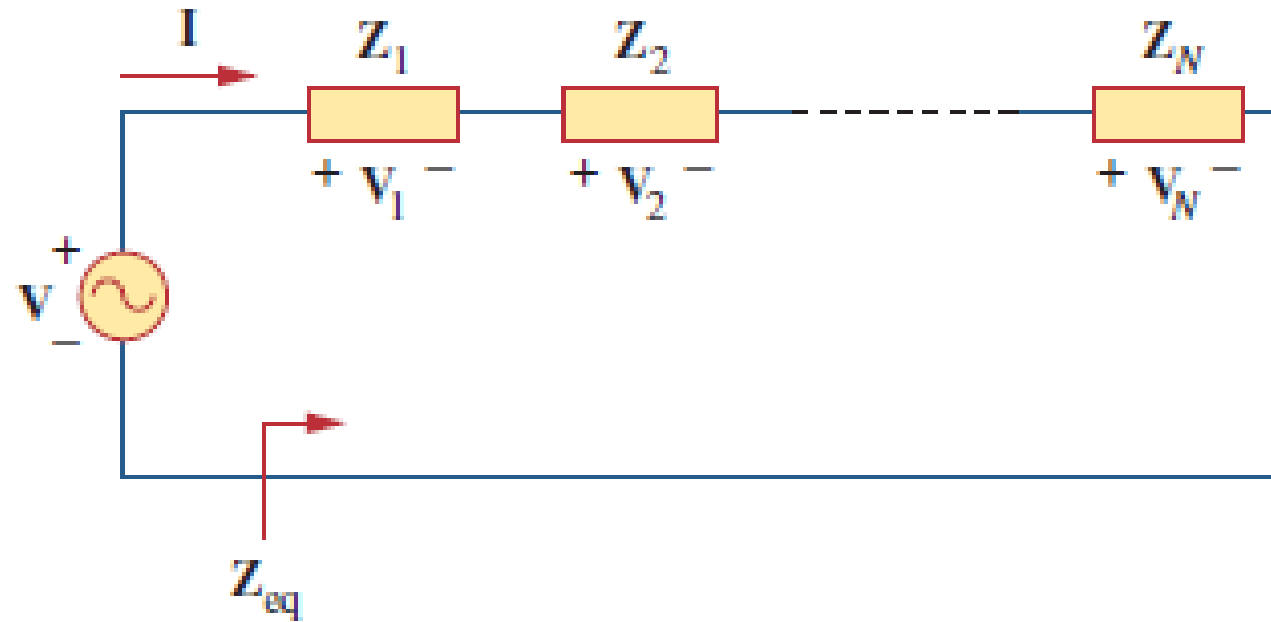
$$V_1 + V_2 + \cdots + V_n = 0$$

- KCL:

Kirchhoff's current law (KCL) states that the algebraic sum of currents entering a node (or a closed boundary) is zero.

$$I_1 + I_2 + \cdots + I_n = 0$$

N Impedances in Series

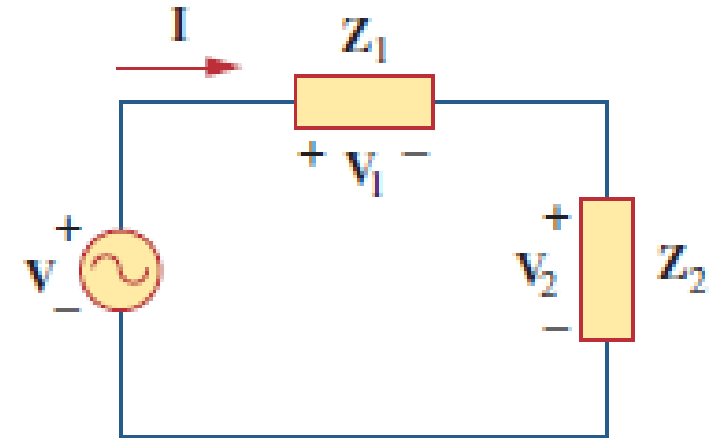


$$Z_{eq} = \frac{V}{I} = Z_1 + Z_2 + \cdots + Z_N$$

Voltage Division

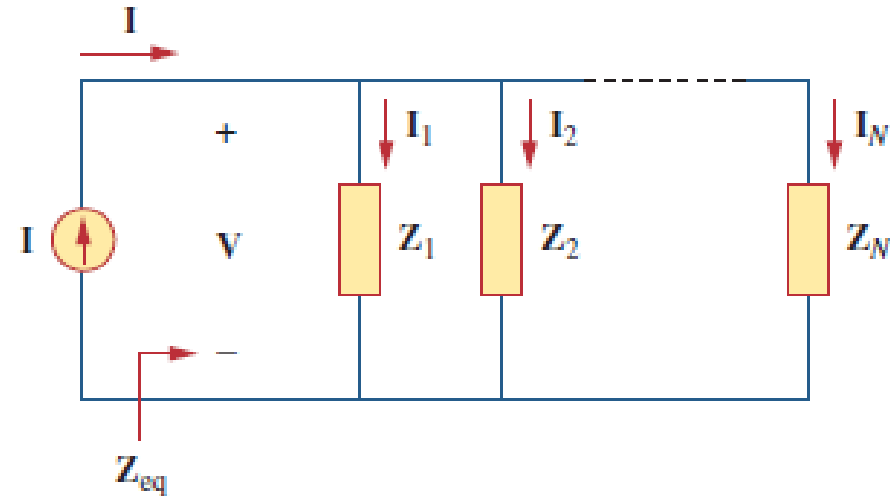
$$I = \frac{V}{Z_1 + Z_2}$$

$$V_1 = Z_1 I \text{ and } V_2 = Z_2 I$$



$$V_1 = \frac{Z_1}{Z_1 + Z_2} V, \quad V_2 = \frac{Z_2}{Z_1 + Z_2} V$$

N Impedances in Parallel



$$I = I_1 + I_2 + \dots + I_N = V \left(\frac{1}{Z_1} + \frac{1}{Z_2} + \dots + \frac{1}{Z_N} \right)$$

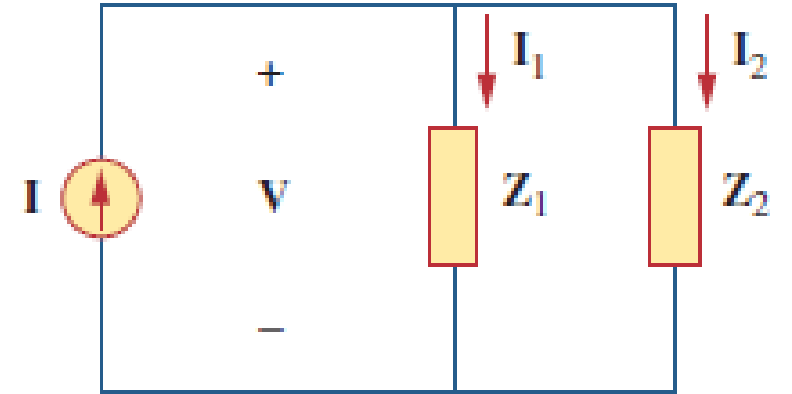
$$\frac{1}{Z_{eq}} = \frac{I}{V} = \frac{1}{Z_1} + \frac{1}{Z_2} + \dots + \frac{1}{Z_N}$$

$$Y_{eq} = Y_1 + Y_2 + \dots + Y_N$$

Current Division

$$Z_{\text{eq}} = \frac{1}{Y_{\text{eq}}} = \frac{1}{Y_1 + Y_2} = \frac{1}{1/Z_1 + 1/Z_2} = \frac{Z_1 Z_2}{Z_1 + Z_2}$$

$$V = I Z_{\text{eq}} = I_1 Z_1 = I_2 Z_2$$



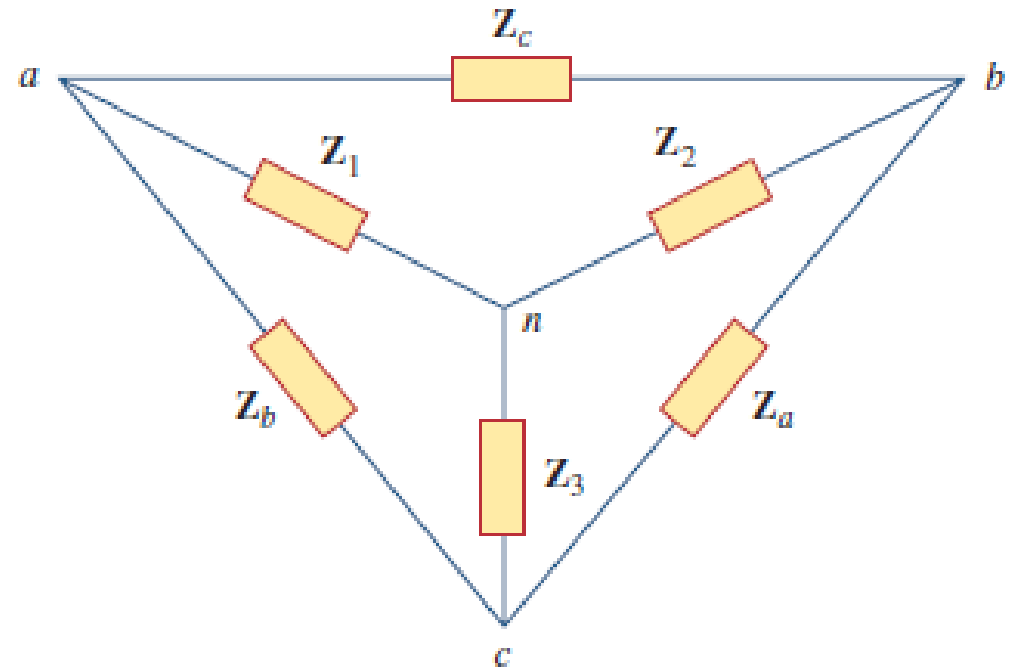
$$I_1 = \frac{Z_2}{Z_1 + Z_2} I, \quad I_2 = \frac{Z_1}{Z_1 + Z_2} I$$

Wye to Delta conversion

$$Z_a = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_1}$$

$$Z_b = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_2}$$

$$Z_c = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_3}$$

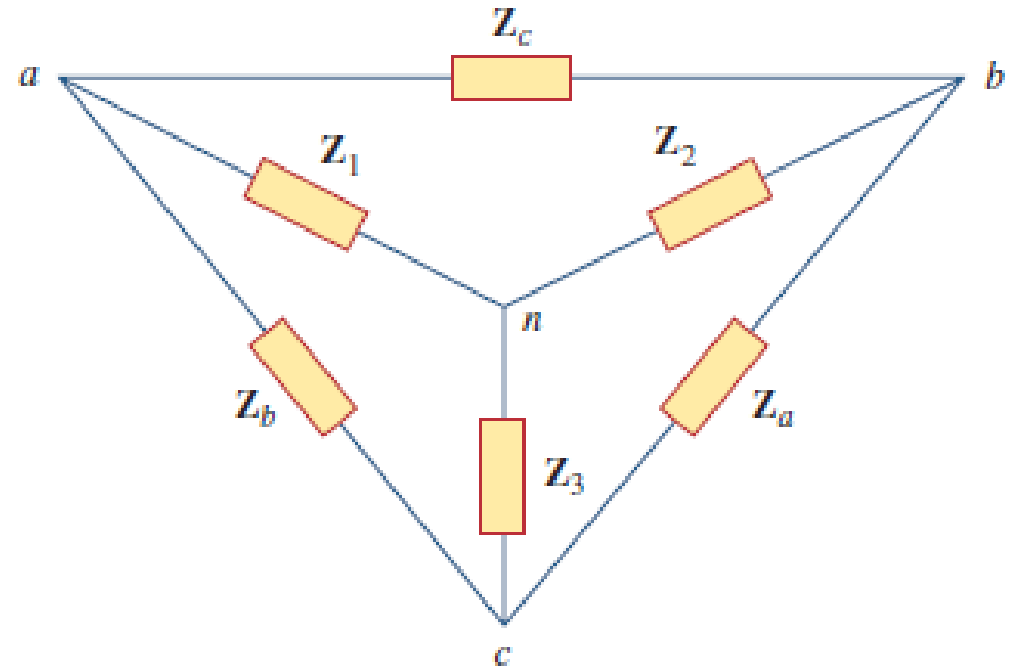


Delta to Wye conversion

$$Z_1 = \frac{Z_b Z_c}{Z_a + Z_b + Z_c}$$

$$Z_2 = \frac{Z_c Z_a}{Z_a + Z_b + Z_c}$$

$$Z_3 = \frac{Z_a Z_b}{Z_a + Z_b + Z_c}$$



A delta or wye circuit is said to be **balanced** if it has equal impedances in all three branches.

$$Z_{\Delta} = 3Z_Y \quad \text{or} \quad Z_Y = \frac{1}{3} Z_{\Delta}$$

where $Z_Y = Z_1 = Z_2 = Z_3$ and $Z_{\Delta} = Z_a = Z_b = Z_c$.

Numerical -1

Find $v(t)$ and $i(t)$ in the circuit shown

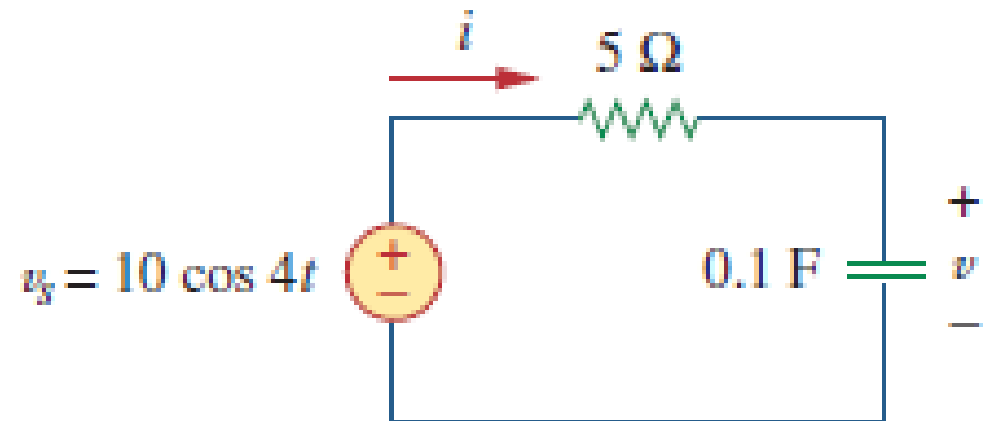
Solution:

From the voltage source $10 \cos 4t$, $\omega = 4$,

$$\mathbf{V}_s = 10 \angle 0^\circ \text{ V}$$

The impedance is

$$\mathbf{Z} = 5 + \frac{1}{j\omega C} = 5 + \frac{1}{j4 \times 0.1} = 5 - j2.5 \, \Omega$$



Hence the current

$$\begin{aligned}\mathbf{I} &= \frac{\mathbf{V}_s}{\mathbf{Z}} = \frac{10\angle 0^\circ}{5 - j2.5} = \frac{10(5 + j2.5)}{5^2 + 2.5^2} \\ &= 1.6 + j0.8 = 1.789\angle 26.57^\circ \text{ A}\end{aligned}$$

The voltage across the capacitor is

$$\begin{aligned}\mathbf{V} &= \mathbf{I}Z_C = \frac{\mathbf{I}}{j\omega C} = \frac{1.789\angle 26.57^\circ}{j4 \times 0.1} \\ &= \frac{1.789\angle 26.57^\circ}{0.4\angle 90^\circ} = 4.47\angle -63.43^\circ \text{ V}\end{aligned}$$

$$i(t) = 1.789 \cos(4t + 26.57^\circ) \text{ A}$$

$$v(t) = 4.47 \cos(4t - 63.43^\circ) \text{ V}$$

$i(t)$ leads $v(t)$ by 90° as expected

Numerical -2

Find the input impedance of the circuit

Assume that the circuit operates at $\omega = 50$ rad/s.

Solution:

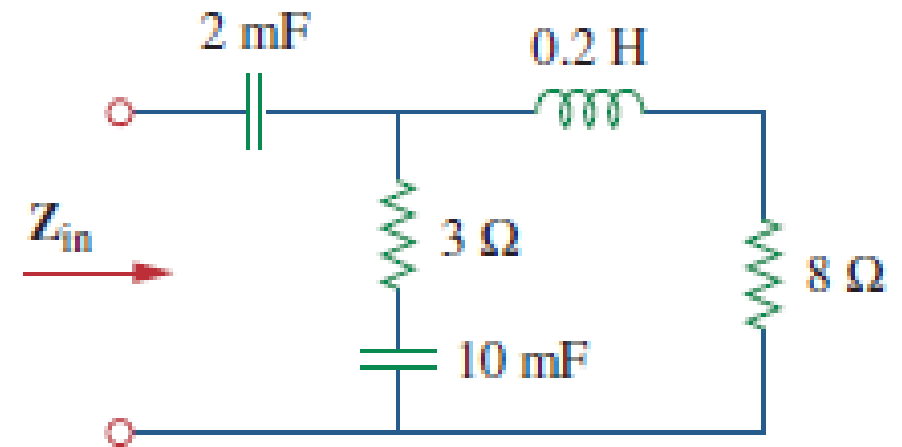
Let

Z_1 = Impedance of the 2-mF capacitor

Z_2 = Impedance of the 3- Ω resistor in series with the 10-mF capacitor

Z_3 = Impedance of the 0.2-H inductor in series with the 8- Ω resistor

$$Z_1 = \frac{1}{j\omega C} = \frac{1}{j50 \times 2 \times 10^{-3}} = -j10 \Omega$$



$$\mathbf{Z}_2 = 3 + \frac{1}{j\omega C} = 3 + \frac{1}{j50 \times 10 \times 10^{-3}} = (3 - j2) \, \Omega$$

$$\mathbf{Z}_3 = 8 + j\omega L = 8 + j50 \times 0.2 = (8 + j10) \, \Omega$$

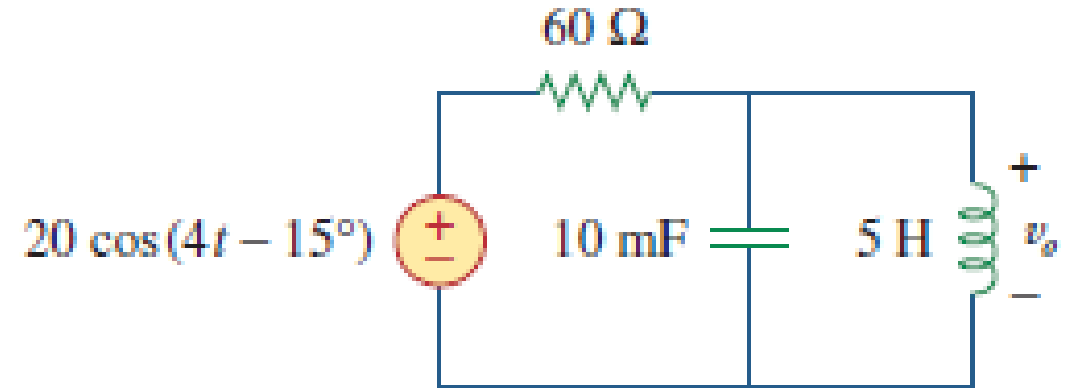
$$\begin{aligned} \mathbf{Z}_{\text{in}} &= \mathbf{Z}_1 + \mathbf{Z}_2 \parallel \mathbf{Z}_3 = -j10 + \frac{(3 - j2)(8 + j10)}{11 + j8} \\ &= -j10 + \frac{(44 + j14)(11 - j8)}{11^2 + 8^2} = -j10 + 3.22 - j1.07 \, \Omega \end{aligned}$$

$$\mathbf{Z}_{\text{in}} = 3.22 - j11.07 \, \Omega$$

Numerical -3

Determine $v_o(t)$ in the circuit

Solution:



$$v_s = 20 \cos(4t - 15^\circ) \quad \Rightarrow \quad \mathbf{V}_s = 20 \angle -15^\circ \text{ V}, \quad \omega = 4$$

$$10 \text{ mF} \quad \Rightarrow \quad \frac{1}{j\omega C} = \frac{1}{j4 \times 10 \times 10^{-3}} \\ = -j25 \Omega$$

$$5 \text{ H} \quad \Rightarrow \quad j\omega L = j4 \times 5 = j20 \Omega$$

Z_1 = Impedance of the 60- Ω resistor

Z_2 = Impedance of the parallel combination of the 10-mF capacitor and the 5-H inductor

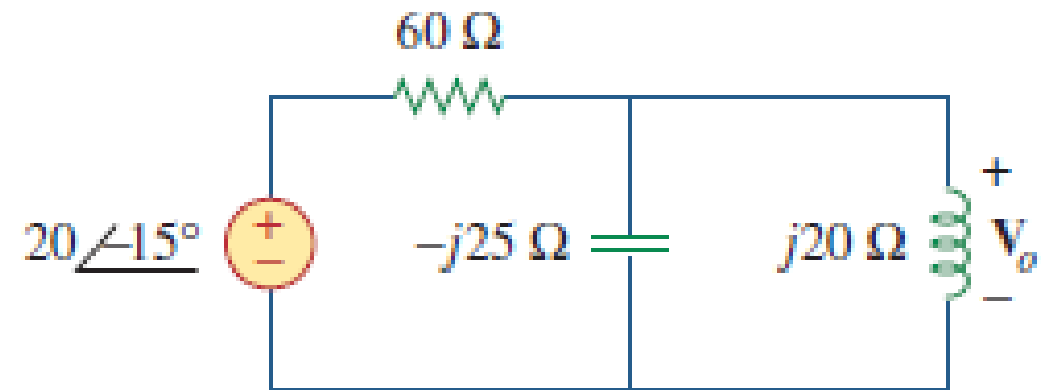
$$Z_1 = 60 \, \Omega$$

$$Z_2 = -j25 \parallel j20 = \frac{-j25 \times j20}{-j25 + j20} = j100 \, \Omega$$

By the voltage-division principle,

$$\begin{aligned} V_o &= \frac{Z_2}{Z_1 + Z_2} V_s = \frac{j100}{60 + j100} (20 \angle -15^\circ) \\ &= (0.8575 \angle 30.96^\circ) (20 \angle -15^\circ) = 17.15 \angle 15.96^\circ \text{ V} \end{aligned}$$

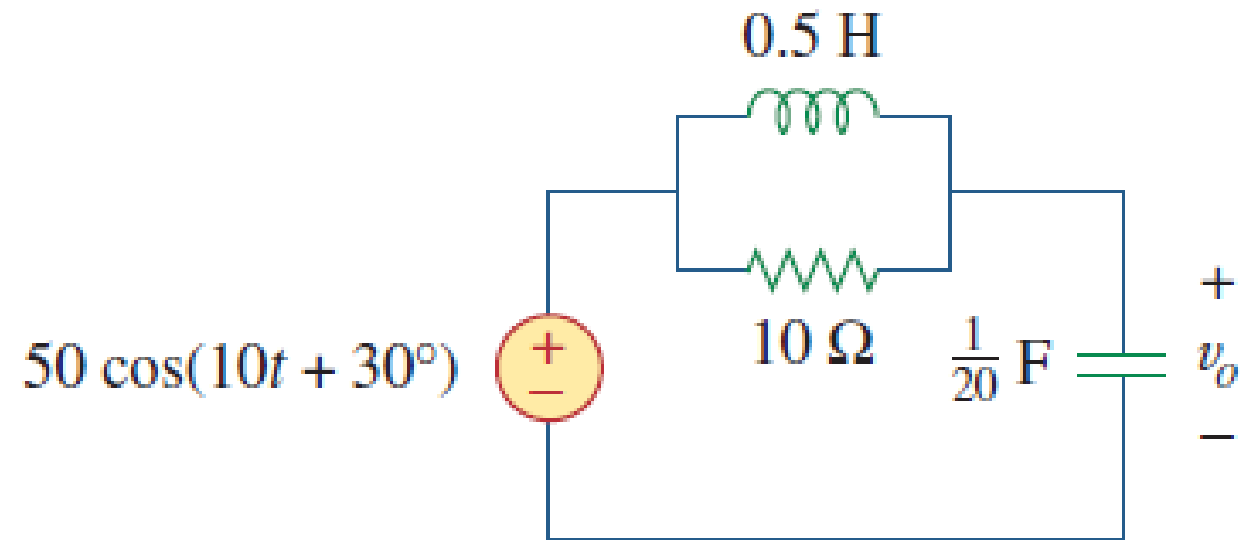
$$v_o(t) = 17.15 \cos(4t + 15.96^\circ) \text{ V}$$



The frequency domain equivalent of the circuit

Numerical -4

Determine $v_o(t)$ in the circuit



Answer: $v_o(t) = 35.36 \cos(10t - 105^\circ) \text{ V}.$